



METHADONE SYRUP 5MG/ML

Each 5ml contains:
Methadone HCl 25mg

Product Description: A red colour syrup without flavour.

Pharmacodynamics:

Methadone is a strong opioid agonist with actions predominantly at the μ receptor. The analgesic activity of the racemate is almost entirely due to the δ -isomer, which is at least 10 times more potent as an analgesic than the δ -isomer. The δ -isomer lacks significant respiratory depressant activity but does have anti-tussive effects. Methadone also has some agonist actions at the κ and δ opiate receptors. These actions result in analgesia, depression of respiration, suppression of cough, nausea and vomiting (via an effect of the chemoreceptor trigger zone) and constipation. An effect on the nucleus of the oculomotor nerve, and perhaps on opioid receptors in the pupillary muscles, causes pupillary constriction. All these effects are reversible by naloxone. Like many basic drugs methadone enters most cells and releases histamine by a non-immunological mechanism. It causes a dependence syndrome of the morphine type.

Pharmacokinetics:

Methadone is well absorbed from the gastrointestinal tract, but undergoes fairly extensive first pass metabolism. It is bound to albumin and other plasma proteins and to tissue proteins, the concentrations in the lung, liver and kidneys being much higher than in blood. With regular oral doses of 100–120mg daily, plasma concentrations rise from trough levels of approximately 500µg/L to a peak of about 900µg/L in 4 hours. Marked variations in plasma levels occur in dependent persons on a stable dose of oral methadone, without any relation to symptoms. Methadone is secreted in sweat and found in saliva and in high concentrations in gastric juice.

Indication:

For use in the treatment of opioid drug addictions (as a narcotic abstinence syndrome suppressant).

Recommended Dosage:

Adults: Initially 10–20 mg per day, increasing by 10–20 mg per day until there are no signs of withdrawal or intoxication. The usual dose is 40–60mg per day. The dose is adjusted according to the degree of dependence with the aim of gradual reduction.

Route of Administration:

Oral

Contraindications:

Respiratory depression, obstructive airways disease, raised intracranial pressure or head injury. Concurrent administration with MAO inhibitors or within 2 weeks of discontinuation of treatment with them. Patients dependent on non-opioid drugs. Use during an asthma attack is not recommended. Known hypersensitivity to the hydroxyzines. Use during labour is not recommended, the prolonged duration of action increases the risk of neonatal depression. Methadone is not suitable for children.

Warnings and Precautions:

Caution should be exercised in patients with hepatic or renal dysfunction, hypothyroidism or prostatic hypertrophy. In the case of the elderly or ill patients repeated dose should only be given with extreme caution. As with other opioids, methadone may cause troublesome constipation, which is particularly dangerous in patients with severe hepatic impairment, and measures to avoid constipation should be initiated early. Cases of QT interval prolongation and torsades de pointes have been reported during treatment with methadone, particularly at high doses > 100mg/d. Methadone should be administered with caution to patients at risk for development of prolonged QT interval, e.g. in case of history of cardiac conduction abnormalities, advanced heart disease or ischaemic heart disease, liver disease, family history of sudden death, electrolyte abnormalities (i.e. hypokalaemia, hypomagnesaemia), concomitant treatment with drugs that have a potential for QT-prolongation, concomitant treatment with drugs which may cause electrolyte abnormalities, concomitant treatment with cytochrome P450 CYP3A4 inhibitors (see Interactions with Other Medicaments). In patients with recognised risk factors for QT-prolongation, or in case of concomitant treatment with drugs that have a potential for QT-prolongation, ECG monitoring is recommended prior to methadone treatment, with a further ECG test at dose stabilisation. ECG monitoring is recommended, in patients without recognised risk factors for QT-prolongation, before dose titration above 100mg/d and at seven days after titration.

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of SW Methadone Syrup with serotonergic drugs (see Interactions with Other Medicaments). This may occur within the recommended dosage range. Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue SW Methadone Syrup if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. When the patient is off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience).

Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of SW Methadone Syrup with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use. If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when SW Methadone Syrup is used with benzodiazepines. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See section Interactions with Other Medicaments).

Interactions with Other Medicaments:

Alcohol: May induce serious respiratory depression and hypotension.

Rifampin: Reduced opiate effect due to increased metabolism.

Anticonvulsants such as Phenytoin, Phenobarbital, Carbamazepine and Primidone: Induces methadone metabolism with the risk of precipitating withdrawal syndrome.

Serotonergic Drugs:

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue SW Methadone Syrup if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

Antiretroviral Agents such as Nevirapine, Efavirenz, Nelfinavir, Ritonavir:

These agents may decrease plasma concentrations of methadone by increasing its hepatic metabolism. Methadone may increase the plasma concentration of zidovudine. Narcotic withdrawal syndrome has been reported in patients treated with some retroviral agents and methadone concomitantly. Methadone maintained patients beginning antiretroviral therapy should be monitored for evidence of withdrawal and methadone dose should be adjusted accordingly.

Cyflousofin:

Concomitant use may lead to sedation, confusion and respiratory depression.

Cytochrome P450 3A4 inhibitors:

Methadone clearance is decreased when co-administered with drugs which inhibit CYP3A4 activity, such as some anti-HIV agents, macrolide antibiotics, cimetidine and azole antifungal agents (since the metabolism of methadone is mediated by the CYP3A4 isoenzyme). Potentiation of opiate action with the risk of methadone toxicity. In patients taking drugs affecting cardiac conduction, or drugs which may affect electrolyte balance there is a risk of cardiac events when methadone is taken concurrently.

Benzodiazepines:

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death. The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABA_A sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists. Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions). Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Pregnancy and Lactation:

There is no, or inadequate, evidence of safety in human pregnancy but the drug has been widely used for many years without apparent ill-consequence and animal studies have not shown any hazard. It may be necessary to increase the dose of methadone if withdrawal symptoms develop. Increased clearance and reduced plasma levels have been reported during pregnancy. It should not be used during labour (see Contraindications). Methadone is excreted in breast milk. Breast feeding is not recommended during methadone treatment.

Side Effects:

Postmarketing Experience:

Metabolic and nutritional disorders

Frequency "not known": hypoglycaemia

Serotonin syndrome

(See Warnings and Precautions)

Adrenal insufficiency

(See Warnings and Precautions)

Androgen deficiency

Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility

Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible. Long-term administration may produce excessive sweating and raised prolactin levels. Cases of QT prolongation and torsades de pointes have been rarely reported.

Symptoms and Treatment of Overdose:

Symptoms: Serious overdose is characterised by respiratory depression, extreme somnolence progressing to stupor or coma, maximally constricted pupils, skeletal muscle flaccidity, cold and clammy skin and sometimes bradycardia and hypotension. In severe overdose, particularly by the intravenous route, apnea, circulatory collapse, cardiac arrest and death may occur.

Treatment: A patent airway and assisted or controlled ventilation must be assured.

Narcotic antagonist may be required, but it should be remembered that methadone is a long-acting depressant (36 to 48 hours), whereas antagonists act for 1 to 3 hours, so that treatment with the latter must be repeated as needed. An antagonist should not be administered, however, in the absence of clinically significant respiratory or cardiovascular depression. The administration of naloxone is advised.

Oxygen, intravenous fluids, vasopressors and other supportive measures should be employed as indicated. In a person physically dependent on narcotics, administration of the usual dose of a narcotic antagonist will precipitate an acute withdrawal syndrome: use of the antagonist in such a person should be avoided if possible but if it must be used to treat serious respiratory depression it should be administered with great care.

Effects on Ability to Drive and Use Machine:

Ability to drive and use machines may be severely affected during and after treatment with methadone. The time after which such activities may be safely resumed is extremely patient dependent and must be decided by the physician. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined.

Storage Conditions: Store at or below 30°C. Protect from light.

Pack size: A bottle of 60ml, 100ml and 120ml.

Pack size (export only) : A bottle of 500ml

Pack size (export/tender) : A bottle of 1 litre.

Shelf-life: 3 years.

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN/LOCAL GENERAL PRACTITIONER/PHARMACIST.

Manufacturer & Product Registration Holder:

Sunward Pharmaceutical Sdn. Bhd.

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